

EZNX-ATLAS-A: Measuring the Incremental Contribution of Clinical Metadata to 12-Lead ECG Superclass Classification on PTB-XL

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Abstract

We present EZNX-ATLAS-A, a 3.95 M-parameter dual-branch model for paired multi-seed quantification of the incremental contribution of clinical metadata to 12-lead ECG superclass classification on PTB-XL 1.0.3. The architecture couples a residual ECG encoder with a demographic-and-anthropometric branch, an explicit metadata-availability score (q_{meta}), and GLU-inspired late fusion. Confirmatory evidence is provided by Group A of a pre-declared **250-run** campaign: three metadata variants $\times$ **20 seeds**, paired Wilcoxon signed-rank tests with Benjamini–Hochberg correction (**pre-specified 3-test family**). Two statistically significant gains are confirmed: demographic variables (age, sex) yield $\Delta = +0.0007$ macro-AUC (95% paired CI [0.0002, 0.0011], $p_{\text{BH}} \approx 0.009$); full metadata yield $\Delta = +0.0019$ macro-AUC (95% paired CI [0.0014, 0.0023], $p_{\text{BH}} < 0.001$). Post-hoc per-class decomposition localises the demographic gain to NORM and MI and the full-metadata gain to NORM, MI, and STTC; CD and HYP show no significant effect. Exploratory groups (B–F) show no hyperparameter sensitivity and reveal fold-level absolute-AUC variation. The absolute gains are small ($\Delta < 0.002$) and should not be interpreted as evidence of clinical utility without external validation. These findings constitute a statistically principled, reproducible baseline for multimodal ECG evaluation and demonstrate the value of a pre-specified seed-matched measurement framework with explicit multiplicity control.

Keywords: electrocardiogram; multimodal fusion; Wilcoxon signed-rank test; missing data; PTB-XL; myocardial infarction; reproducibility; nonparametric inference

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1. Introduction

Automatic interpretation of the 12-lead electrocardiogram is one of the most mature applications of clinical deep learning. Hannun *et al.* [1] reported cardiologist-level arrhythmia detection on ambulatory single-lead recordings; Ribeiro *et al.* [2] demonstrated

large-scale automatic diagnosis from standard 12-lead ECGs; and Attia *et al.* [3] showed that a 12-lead CNN could screen for reduced left-ventricular ejection fraction. Siontis *et al.* [4] review these and related AI-ECG applications in cardiovascular disease management. On the public PTB-XL corpus [5], the canonical benchmark of Strodthoff *et al.* [6] places superclass macro-AUC at 0.928 ± 0.005 for a 1D-ResNet. More recent self-supervised systems [7–10] improve selected ECG benchmarks, but they are waveform-centric and therefore do not directly answer how much structured clinical metadata adds to PTB-XL superclass classification.

A natural next step is *multimodal fusion* with structured clinical metadata, since PTB-XL exposes age and sex for nearly all records and height/weight-derived anthropometrics for a smaller subset. Recent ECG-fusion works [11–14] show that ECG representations can be combined with clinical variables, patient-characteristic modules, or engineered feature tables. However, our paper-by-paper audit (Tables 9–10; not a systematic PRISMA review—papers selected as the most directly comparable to our design) found that these studies differ in task, dataset, and metadata definition; their reported gains should therefore not be read as direct numerical baselines for PTB-XL superclass demographic/anthropometric fusion. The specific gap we target is narrower: seed-matched ablation of demographics versus anthropometrics, per-class decomposition, and inference-time missingness stress testing.

We contribute a targeted remedy. Rather than proposing yet another ECG backbone, we design **EZNX-ATLAS-A**—a deliberately compact (3.95 M parameters) quality-gated dual-branch architecture—and use it as an *instrument* to address six questions. We intentionally keep a compact residual backbone, rather than escalate simultaneously to a larger Transformer or foundation waveform encoder, so that measured gains are easier to attribute to the metadata pathway itself and so that 250 unique runs across six experimental groups remain computationally tractable:

- Q1. When, if ever, does structured clinical metadata detectably improve PTB-XL superclass classification, once seed variance is accounted for?
- Q2. Which *type* of metadata (demographic vs. anthropometric) helps which pathology class?
- Q3. Can a quality-gated fusion degrade gracefully as anthropometric metadata fields are withheld, remaining near the ECG-only and demographics-only baselines?
- Q4. How do our measured effects compare with recent multimodal ECG reports once task, dataset, metadata definition, and statistical protocol are audited?
- Q5. Are the reported gains robust to architectural variants (fusion mechanism, gating, residual path) and is ECG signal necessary?
- Q6. Are the results specific to fold 10, or do they generalise across alternative data splits?

Our answers, detailed in Sections 6–7, are: (Q1) *both* demographic-only and full-metadata variants show statistically significant gains with **20 paired seeds**: demographics alone (demo minus none) yield $\Delta = +0.0007$ macro-AUC (95% paired CI [0.0002, 0.0011], $p_{\text{BH}} \approx 0.009$), while full metadata (demo+anthro minus none) yield $\Delta = +0.0019$ macro-AUC (95% paired CI [0.0014, 0.0023], $p_{\text{BH}} < 0.001$). (Q2) The class-wise effect of full metadata concentrates in NORM, MI and STTC, while HYP shows no significant gain; (Q3) macro-AUC loss ≈ 0.0014 under full anthropometric masking at inference, converging to the demographics-only baseline; (Q4) the audited literature motivates multimodal fusion but does not provide a directly comparable seed-matched PTB-XL demographic/anthropometric baseline. (Q5) Exploratory Group E ablations (8 architectural

variants, 10 seeds each, uncorrected) suggest that the ECG signal is indispensable (meta-only macro-AUC 0.6966, $p=0.002$, exploratory) while the specific fusion mechanism is not a driver of performance differences. (Q6) Group F multi-split analysis (4 alternative folds, 5 seeds each) shows that fold 10 lies below the inter-fold range [0.9400, 0.9445] for the demo+anthro condition, indicating that fold 10 yields lower absolute full-metadata performance than the tested alternatives; within-fold paired comparisons are unaffected. These small but reproducible effects support a cautious measurement study replacing broad multimodal claims with statistically honest, class-localised ones.

Beyond the ECG application, this paper introduces a reusable statistical measurement pipeline for multimodal-ML ablations: exact two-sided paired Wilcoxon signed-rank tests on seed-matched differences, Benjamini–Hochberg FDR control at $q=0.05$ across a **pre-declared 3-test family** (the three pairwise macro-AUC contrasts of Group A), and percentile bootstrap CIs for paired differences. This pipeline applies to any ablation study in which model variants are compared across repeated random seeds.

2. Related Work

2.1. Supervised benchmarks on PTB-XL

Wagner *et al.* [5] released PTB-XL, a 21 799-record 12-lead corpus with SCP-ECG multi-label annotations aggregated into 5 superclasses, 23 sub-classes and 44 diagnostic statements, together with a stratified, patient-disjoint 10-fold partition. Strodtzoff *et al.* [6] established the canonical benchmark, reporting 10-fold cross-validated macro-AUC 0.928 ± 0.005 on super-diag with xresnet1d101. Nonaka and Seita [15] swept nine 1D-CNN backbones and found ResNet-18 competitive with deeper variants. The Bydgoszcz group [16,17] studied entropy features and few-shot learning but used a non-standard 70/15/15 split, making their numbers incomparable to the canonical benchmark. In the supervised PTB-XL benchmark papers audited here, we did not identify a seed-matched age/sex vs. anthropometric ablation with an explicit missing-metadata stress test.

2.2. Landmark clinical ECG models

Hannun *et al.* [1] reached cardiologist-level rhythm classification on 91 232 single-lead ambulatory ECGs; Ribeiro *et al.* [2] generalised to 12-lead short recordings on a 2M-record Brazilian telehealth corpus. Attia *et al.* [3] showed a CNN could screen for LV ejection fraction $\leq 35\%$ directly from the 12-lead ECG (AUC 0.93). Critically for the present paper, Attia *et al.* [18] demonstrated that age and sex are themselves *recoverable* from the raw ECG (age MAE 6.9 y, sex AUC 0.97), supporting the idea that demographic signal is already partly present in the waveform. Siontis *et al.* [4] identify this latent-demographic phenomenon as both an opportunity and a confounding risk.

2.3. Self-supervised and foundation ECG models

Contrastive and masked-modelling approaches—CLOCS [19], CPC-based pretraining [7], ST-MEM [20], the ECG joint-embedding predictive architecture (ECG-JEPA) [8], the HuBERT-ECG preprint [9], and the ACM MM 2025 TolerantECG model [10]—study representation learning and robustness from waveform inputs alone, including PTB-XL downstream evaluations in some cases. They are therefore adjacent to, but distinct from, the metadata-value question studied here. The preprint and conference works in this paragraph are cited as contextual neighbouring work rather than as pre-specified baselines.

2.4. Multimodal ECG + tabular fusion

Explicit ECG + tabular or feature-fusion architectures have begun to appear, but they are not interchangeable baselines. Qiu *et al.* [11] fuse a transformer ECG encoder with

clinical features for atrial-fibrillation recurrence prediction after ablation, outside PTB-XL. The preprint rECGnition_v2.0 system [12] studies self-attentive canonical fusion on MIT-BIH, INCARTDB and the European ST-T Database. Among PTB-XL-family studies, Atwa *et al.* [13] (whose published title uses the abbreviation “PTB-X”, referring to the same PhysioNet corpus) is the closest published demographic-fusion comparator because it adds age and sex to CNN/VGG-style arrhythmia models; however, it reports accuracy-oriented results across 2-, 5-, 10-, and 15-class tasks with augmentation rather than seed-matched macro-AUC on the canonical five-superclass split. In parallel, PTB-XL+ [21] augments PTB-XL with engineered Uni-G, 12SL and ECGDeli feature tables; Holloway *et al.* [14] compare raw PTB-XL signals with such PTB-XL+ feature representations. These feature tables are derived from the ECG itself rather than from EHR-like anthropometrics. We therefore treat these papers as motivation and methodological context, not as direct numerical comparators; the exact comparability audit is summarised in Tables 9–10.

2.5. Gated fusion mechanisms

The Gated Linear Unit (GLU) of Dauphin *et al.* [22] provides a simple multiplicative gate $\mathbf{h} \odot \sigma(\cdot)$ that preserves a direct linear gradient path through the ungated branch. The Gated Multimodal Unit [23] extends this to modality-level mixing. In medical imaging, Chen *et al.* [24] use feature disentanglement plus gated fusion for brain-tumor segmentation, and Kim *et al.* [25] introduce reliability-aware gating for degraded or low-quality modalities. We use a GLU-based gate conditioned on an explicit scalar quality score q_{meta} derived from the metadata-availability mask.

2.6. Missing modalities in clinical ML

Baltrušaitis *et al.* [26] survey the main settings of multimodal learning, including fusion and co-learning under incomplete modalities. HeMIS [27], SMIL [28], and mmFormer [29] address missing modalities in imaging or general multimodal settings rather than ECG+tabular fusion. We study the adjacent ECG+tabular setting by coupling an availability-mask-driven quality score with stochastic sample meta-dropout during training.

2.7. Clinical motivation for anthropometrics in ECG classification

Classical ECG LVH criteria—Sokolow–Lyon [30], Cornell voltage/duration product [31], Romhilt–Estes [32], and Peguero–Lo Presti [33]—depend on measured voltage and are affected by body habitus. Left ventricular mass is itself routinely indexed to body surface area or height^{2.7}. *A priori*, these cardiology facts suggest that the HYP superclass could benefit from anthropometric metadata. More broadly, ECG amplitudes and repolarisation measurements relevant to MI and STTC can also be influenced by torso conduction properties, lead geometry, and recording context [34,35], suggesting possible relevance of body-size priors across morphology-rich classes. Our per-class results in Section 6 are compatible with this broader relevance while not supporting a strong HYP-specific effect at the superclass level.

3. Dataset and Preprocessing

3.1. PTB-XL 1.0.3

We use PTB-XL version 1.0.3 on PhysioNet [5,36] at 100 Hz sampling rate ($T=1000$ samples, 10 s, 12 leads). Following [6], SCP-ECG diagnostic statements are mapped to the 5-class super-diagnostic task {NORM, MI, STTC, CD, HYP} using `scp_statements.csv`. In the implemented loader, a record is positive for a superclass if any diagnostic SCP code assigned to that class is present in `scp_codes`; no additional likelihood threshold is

applied at load time, which is consistent with the standard superdiagnostic benchmark label protocol [6]. The official stratified-by-label and patient-disjoint 10-fold partition is used unchanged in the working index: folds 1–8 for training, fold 9 for validation, fold-10 for test, yielding 17,418 / 2,183 / 2,198 records, respectively (Table 5). These fold counts sum to 21 799 records, matching the current PhysioNet v1.0.3 release after the duplicate removals documented in its release notes.¹

3.2. Formal multi-label problem setting

For each record i , the loader returns a quadruple

$$(\mathbf{X}_i, \mathbf{x}_i, \mathbf{m}_i, \mathbf{y}_i) \in \mathbb{R}^{12 \times 1000} \times \mathbb{R}^8 \times \{0, 1\}^8 \times \{0, 1\}^5,$$

where $\mathbf{X}_i$ is the ECG, $\mathbf{x}_i$ is the imputed metadata vector, $\mathbf{m}_i$ is its availability mask, and $\mathbf{y}_i$ is the multi-hot superclass target. A model with parameters θ emits logits $\ell_i(\theta) \in \mathbb{R}^5$ and probabilities $\mathbf{p}_i = \sigma(\ell_i)$. The principal metric is

$$\text{macroAUC}(\theta) = \frac{1}{5} \sum_{c=1}^5 \text{AUC}(\{(y_{ic}, p_{ic})\}_{i \in \mathcal{D}_{\text{test}}}),$$

with class-wise AUC skipped only if a class is degenerate, which does not occur for the PTB-XL superclasses on fold-10. The ablations below are deterministic interventions on $(\mathbf{x}_i, \mathbf{m}_i)$, so seed-matched comparisons estimate the incremental contribution of metadata while holding the architecture, optimiser, folds, and random seed fixed.

3.3. Metadata extraction and normalisation

From the PTB-XL metadata we construct the 8-dimensional tabular vector

$$\mathbf{x}_{\text{meta}} = \left[\underbrace{\text{age}_z, \text{sex01}}_{\text{demographic}}, \underbrace{\text{height}_z, \text{weight}_z, \text{bmi}_z}_{\text{anthropometric}}, \underbrace{m_h, m_w, m_{\text{bmi}}}_{\text{missingness indicators}} \right] \in \mathbb{R}^8, \quad (1)$$

where $\text{sex01} \in \{0, 1\}$ ($\text{sex01} = 1$ denotes male, $\text{sex01} = 0$ denotes female, following the PTB-XL encoding), subscript z denotes standardisation using training-fold mean and population standard deviation ($\text{ddof}=0$; minimal difference for $n_{\text{train}}=17,418$) after median imputation, BMI is derived from height and weight when both are observed and otherwise imputed, and $m \in \{0, 1\}$ is a derived missingness flag set to 1 if the corresponding anthropometric field is absent (0 if originally observed); note that this is opposite to the availability mask $\mathbf{m}$ described below, which uses 1 to indicate presence. $\mathbf{x}_{\text{meta}}$ therefore bundles both imputed feature values and the three derived missingness indicators into the same 8-dimensional vector. The separate availability mask $\mathbf{m} \in \{0, 1\}^8$ (index columns `mask_*`) flags whether each entry of $\mathbf{x}_{\text{meta}}$ was originally present; because m_h, m_w, m_{bmi} are derived binary constants, their availability mask bits mirror the corresponding anthropometric field's availability (`mask_miss_height=mask_height`, etc.): the mask bit is 1 when the field is originally observed and 0 when the field is absent. Consequently, a missingness indicator carries its availability mask at 0 precisely when the field is absent (i.e., when the indicator is most informative); the primary missing-data robustness mechanism is therefore sample meta-dropout and median imputation rather than attention via missingness indicator availability (see also Discussion Q3, Section 7).

¹ PTB-XL v1.0.3 release notes are available at <https://physionet.org/content/ptb-xl/1.0.3/>; the 21,799 record count is verified by loading the `ptbx1_database.csv` index and applying the standard stratified fold assignment.

3.4. Positive class weights from training-set prevalence

Let $n_{\text{pos},c}$ and $n_{\text{neg},c}$ denote the numbers of positive and negative training-fold records for superclass c , and let $f_c = n_{\text{pos},c} / (n_{\text{pos},c} + n_{\text{neg},c})$ be its prevalence. We derive the binary cross-entropy positive weights as the clipped negative-to-positive ratio:

$$w_c^+ = \text{clip}\left(\frac{n_{\text{neg},c}}{n_{\text{pos},c}}, 0.5, 30.0\right), \quad c \in \{\text{NORM}, \text{MI}, \text{STTC}, \text{CD}, \text{HYP}\}, \quad (2)$$

Weights grow with class rarity: the most-prevalent class (NORM, 43.6% of training records) receives the smallest weight, while the rarest (HYP, 12.2%) receives the largest. The resulting weight vector $[w_{\text{NORM}}^+, w_{\text{MI}}^+, w_{\text{STTC}}^+, w_{\text{CD}}^+, w_{\text{HYP}}^+] \approx [1.29, 2.98, 3.16, 3.46, 7.22]$, verified from the logged training outputs, is computed dynamically at the start of each run from the training folds only (never from validation or test folds).

3.5. Preprocessing

Signals are loaded with `wfdb` at 100 Hz, zero-padded at the end if shorter than 1000 samples or truncated from the start if longer, yielding a fixed 1000-sample length; no PTB-XL 100 Hz recording required padding in this study. Signals are then transposed to channel-first shape $(12, T)$, cast to `float32`, and divided by 5.0 before entering the model. Metadata normalisation (z-scoring) and out-of-range conversion to missing are performed during index construction (before training): age 0–120 years (PTB-XL records with age >89 may be encoded as elevated values under HIPAA de-identification safe harbour; such values exceed 120 and are treated as missing), height 120–210 cm, weight 30–250 kg, and BMI 10–60; the training loader reads from the resulting precomputed index. Evaluation splits apply only the fixed divisor; training additionally uses light stochastic augmentation. The augmentation block fires with probability 0.5 per training mini-batch; inside it, three operations each trigger independently at probability 0.3: additive Gaussian noise ($\sigma=0.02$), a random integer temporal shift (uniform on $[-20, +19]$ samples), and a multiplicative amplitude jitter (uniform from $[0.95, 1.05]$).

4. Model Architecture: EZNX-ATLAS-A

Figure 1 shows the complete architecture. We describe each block with its dimensional signature and design rationale.

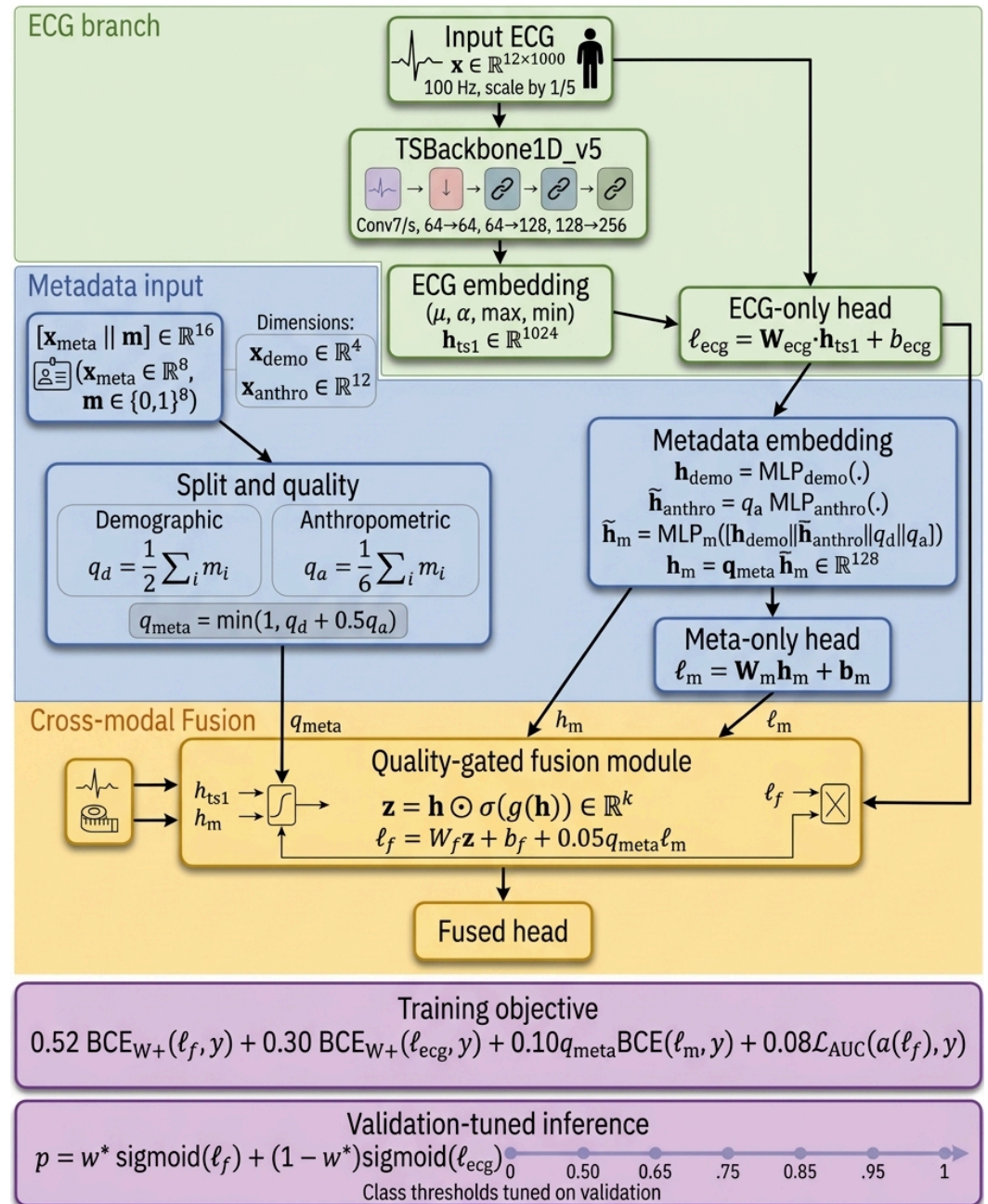


Figure 1. EZNX-ATLAS-A architecture. The 1D residual ECG backbone emits a 1024-dim temporal-statistics embedding $\mathbf{h}_{ts1}$. A dual demographic/anthropometric MLP stack produces $\mathbf{h}_m \in \mathbb{R}^{128}$, scaled by an explicit availability score q_{meta} . One zero-initialised 1024×128 residual projection ($\mathbf{W}_{res}$) softly biases the ECG embedding via residual injection (Eq. 8); a second zero-initialised projection ($\mathbf{W}_{addon}$) is instantiated exclusively for the E3 additive-injection ablation (Group E). A GLU-inspired multiplicative gate fuses the concatenated embeddings. Three classification heads (fused, ECG-only, meta-only) enable multi-loss training; inference uses the fused-branch outputs ($w^*=1.0$ fixed a priori in all runs). Table 6 reports the matched parameter budget (3.95 M) and the archived compute provenance.

4.1. ECG branch: TSBBackbone1D_v5

An initial Conv1D ($k=7, s=2, p=3, 12 \rightarrow 64$) with BatchNorm and ReLU is followed by MaxPool ($k=3, s=2, p=1$). Three residual stages, each with two residual blocks, implement the channel schedule $64 \rightarrow 64$ (stride 1), $64 \rightarrow 128$ (stride 2), $128 \rightarrow 256$ (stride 2), giving an

output of shape $256 \times T'$ with $T'=63$ for $T=1000$. A *Temporal Statistics Pool* concatenates mean, standard deviation, max, and min over time:

$$\mathbf{h}_{ts1} = [\mu(\mathbf{F}), \sigma(\mathbf{F}), \max(\mathbf{F}), \min(\mathbf{F})] \in \mathbb{R}^{1024}, \quad \mathbf{F} \in \mathbb{R}^{256 \times T'}. \quad (3)$$

4.2. Dual metadata encoder

The loader returns the 8-dim tabular vector in Eq. (1) together with its 8-dim availability mask $\mathbf{m}$. Each branch receives its feature slice concatenated with the corresponding mask slice, yielding a 4-dim demographic input (2 values + 2 availability bits) and a 12-dim anthropometric input (6 values + 6 availability bits):

$$\mathbf{h}_{\text{demo}} = \text{MLP}_{\text{demo}}(\mathbf{x}_{\text{demo}}) : \mathbb{R}^4 \rightarrow \mathbb{R}^{64} \rightarrow \mathbb{R}^{64} \rightarrow \mathbb{R}^{64}, \quad (4)$$

$$\mathbf{h}_{\text{anthro}} = \text{MLP}_{\text{anthro}}(\mathbf{x}_{\text{anthro}}) : \mathbb{R}^{12} \rightarrow \mathbb{R}^{96} \rightarrow \mathbb{R}^{96} \rightarrow \mathbb{R}^{64}. \quad (5)$$

Each intermediate linear layer uses ReLU, BatchNorm, and 10% Dropout; the final linear layer uses ReLU only.

4.3. Availability-aware quality score

We define per-block availability scores and an aggregate metadata-quality score:

$$q_d = \frac{1}{2} \sum_{i=0}^1 m_i, \quad q_a = \frac{1}{6} \sum_{i=2}^7 m_i, \quad q_{\text{meta}} = \min(1, q_d + 0.5 q_a). \quad (6)$$

In the working index, sex is always observed and age is observed for 98.7% of records, so q_d is usually 1, can drop to 0.5 when age is missing, and reaches 0 only in the theoretical edge case where both are unavailable (not observed in PTB-XL). Consequently, $q_{\text{meta}} = \min(1, q_d + 0.5 q_a) = 1$ for virtually all records in the demo and demo+anthro conditions (since $q_d=1$ implies $q_d + 0.5 q_a \geq 1$). Graceful degradation under anthropometric missingness is therefore driven primarily by the explicit missingness flags (m_h, m_w, m_{bmi}) in $\mathbf{x}_{\text{meta}}$ and by sample meta-dropout; q_{meta} scaling provides the hard gate between the none variant ($q_{\text{meta}}=0$) and any variant with demographics present ($q_{\text{meta}}=1$).

4.4. Meta-fusion MLP with sample meta-dropout

Concatenation of the encoder outputs with the two availability scalars produces:

$$\begin{aligned} \tilde{\mathbf{h}}_{\text{anthro}} &= q_a \mathbf{h}_{\text{anthro}}, \\ \mathbf{u}_m &= [\mathbf{h}_{\text{demo}} \parallel \tilde{\mathbf{h}}_{\text{anthro}} \parallel q_d \parallel q_a] \in \mathbb{R}^{130}, \\ \mathbf{h}_m &= q_{\text{meta}} \text{MLP}_m(\mathbf{u}_m) \in \mathbb{R}^{128}. \end{aligned} \quad (7)$$

During training with probability $p=0.10$, the entire pre-quality embedding is zeroed for the sample (*sample meta-dropout*); at inference the dropout is disabled and q_{meta} scaling is applied.

4.5. Residual injection

$$\mathbf{h}_{ts} = \mathbf{h}_{ts1} + 0.10 q_{\text{meta}} \mathbf{W}_{\text{res}} \mathbf{h}_m, \quad \mathbf{W}_{\text{res}} \in \mathbb{R}^{1024 \times 128}, \mathbf{W}_{\text{res}} \leftarrow \mathbf{0} \text{ at init.} \quad (8)$$

The projection $\mathbf{W}_{\text{res}}$ is zero-initialised, making the injection path neutral at initialisation. A second independent zero-initialised projection $\mathbf{W}_{\text{addon}} \in \mathbb{R}^{1024 \times 128}$ is instantiated exclusively for the E3 additive-injection ablation (Group E; Section 6.7) and is inert in the standard architecture (it receives zero gradient and remains at zero throughout standard-

mode training). Because $\mathbf{h}_m$ is itself scaled by q_{meta} (Eq. (7)), the effective attenuation of the residual path is q_{meta}^2 , aggressively suppressing metadata-to-ECG cross-talk at partial availability.

4.6. GLU-inspired fusion gate

The gate applies element-wise multiplication to the full concatenated embedding (gating applied to the full 1152-dim concatenation without the feature splitting of the original GLU of [22]):

$$\mathbf{h} = [\mathbf{h}_{\text{ts}} \parallel \mathbf{h}_m] \in \mathbb{R}^{1152}, \quad \mathbf{z} = \mathbf{h} \odot \sigma(\text{Linear}_2(\text{ReLU}(\text{Linear}_1(\mathbf{h})))) \in \mathbb{R}^{1152}. \quad (9)$$

4.7. Three heads with meta-correction

$$\ell_{\text{ecg}} = \mathbf{W}_{\text{ecg}} \mathbf{h}_{\text{ts1}} + \mathbf{b}_{\text{ecg}}, \quad (10)$$

$$\ell_{\text{meta}} = \mathbf{W}_{\text{meta}} \mathbf{h}_m + \mathbf{b}_{\text{meta}}, \quad (11)$$

$$\ell_{\text{fused}} = \mathbf{W}_{\text{fused}} \mathbf{z} + \mathbf{b}_{\text{fused}} + 0.05 q_{\text{meta}} \ell_{\text{meta}}. \quad (12)$$

ℓ_{ecg} is computed on $\mathbf{h}_{\text{ts1}}$ before the residual injection in Eq. (8), preserving a strictly unimodal ECG branch. The $0.05 q_{\text{meta}}$ term in Eq. (12) provides a small direct metadata-logit correction to the fused head, gated by availability; at $q_{\text{meta}}=0$ the fused head reduces to $\mathbf{W}_{\text{fused}} \mathbf{z} + \mathbf{b}_{\text{fused}}$ alone.

4.8. Parameter budget

The total is 3.95 M trainable parameters: 0.96 M in the ECG backbone, 0.06 M in the dual metadata encoders and metadata-fusion MLP, 2.66 M in the GLU fusion gate, 0.26 M in the two metadata-to-ECG residual projections ($\mathbf{W}_{\text{res}}$, active in the standard architecture, and $\mathbf{W}_{\text{addon}}$, active in the E3 additive-injection ablation only; each $\mathbb{R}^{1024 \times 128}$, both zero-initialised), and 11.5 k in the three classification heads. The GLU gate accounts for about 67% of trainable parameters.

5. Training Protocol and Experimental Design

5.1. Composite multi-task loss

We minimise

$$\begin{aligned} \mathcal{L} = & 0.52 \text{BCE}_{\mathbf{w}^+}(\ell_{\text{fused}}, \mathbf{y}) + 0.30 \text{BCE}_{\mathbf{w}^+}(\ell_{\text{ecg}}, \mathbf{y}) \\ & + 0.10 q_{\text{meta}} \text{BCE}(\ell_{\text{meta}}, \mathbf{y}) + 0.08 \mathcal{L}_{\text{AUC}}(\sigma(\ell_{\text{fused}}), \mathbf{y}), \end{aligned} \quad (13)$$

where $\text{BCE}_{\mathbf{w}^+}$ is weighted multi-label binary cross-entropy with class weights from Eq. (2) and $\mathcal{L}_{\text{AUC}}$ is an unsquared hinge AUC surrogate pooling all positive/negative label positions across the mini-batch.

5.2. Optimiser and schedule

AdamW with learning rate 10^{-3} , weight decay 5×10^{-4} ; CosineAnnealingWarm-Restarts scheduler with $T_0=10$, $T_{\text{mult}}=2$ (with $T_0=10$ matching the epoch budget, the schedule executes one complete cosine cycle and no warm restart is triggered); batch size 32, gradient accumulation 2 (effective batch 64); gradient clipping at $\|\nabla\|_2 \leq 1.0$; 10 epochs. The saved checkpoint is the epoch with the highest validation macro-AUC of the fused branch (with $w^*=1.0$ fixed). Operating thresholds are tuned per class on validation by maximising F_1 over 61 thresholds from 0.05 to 0.95.

5.3. Inference blending

Per-class probabilities are computed as

$$\mathbf{p}(\mathbf{x}) = w^* \sigma(\ell_{\text{fused}}) + (1 - w^*) \sigma(\ell_{\text{ecg}}), \quad (14)$$

where $w^*=1.0$ is fixed *a priori* according to the pre-declared analysis plan. Test predictions are therefore simply the fused-branch outputs for all runs. No blend-weight grid search is performed; the fused head is used directly.

5.4. Ablation variants

We train three variants that differ *only* in the metadata input:

none $\mathbf{x}_{\text{meta}} \leftarrow \mathbf{0}$, $\mathbf{m} \leftarrow \mathbf{0}$; $q_{\text{meta}}=0$ and the metadata embedding is nulled after quality scaling.
demo age and sex are retained; height, weight, BMI values set to 0, their missingness indicators set to 1, and all anthropometric/missingness masks set to 0.
demo+anthro full 8-dim $\mathbf{x}_{\text{meta}}$ as in Eq. (1).

5.5. Seeds, runs, compute

The full training campaign is organised in **six experimental groups (A–F)** totalling **250 unique runs** (270 seed-variant descriptors, of which 20 are shared between groups):

Group A (primary confirmatory, 60 runs). Three metadata variants (none, demo, demo+anthro) $\times$ **20 seeds** $S=\{2024, \dots, 2043\}$, test fold 10, val fold 9. Python, NumPy, and PyTorch are each re-seeded at the start of every run; the random state is re-initialised immediately before DataLoader construction (after all index and split computations), so that within-seed batch ordering and augmentation trajectories are identical across the three variants for the same seed. In the none variant, $q_{\text{meta}}=0$ nulls all metadata contributions; the metadata encoder parameters receive no gradient and remain at their random initialisation throughout training. Wall-clock time totalled 48.2 h (median 43.8 min/run; range [43.1, 303.5] min, the high value reflecting a transient CPU throttle at epoch 5 with no effect on convergence; the outlier run's final test macro-AUC is within the inter-seed standard deviation of its variant, as confirmed by its seed-level JSON record).

Group B (exploratory, 30 unique runs). Three non-reference meta_hid values $\{32, 64, 256\} \times 10$ seeds, demo+anthro condition; reference meta_hid=128 shares seeds with Group A. Note: meta_hid controls both projection and gate hidden dimensions (entangled).

Group C (exploratory, 40 unique runs). Four non-reference λ_{LAUC} values $\{0.00, 0.04, 0.12, 0.16\} \times 10$ seeds; reference $\lambda=0.08$ shares seeds with Group A. Note: $w_{\text{fused}} = \max(0, 0.60 - \lambda_{\text{LAUC}})$, so both weights change together. Additionally, the code normalises all four loss weights by their sum (total_w) to ensure a bounded effective loss; for the values tested, total_w ranges from 0.98 ($\lambda=0.16$) to 1.00 ($\lambda=0.08$), making the renormalisation effect marginal but non-zero.

Group D (exploratory, 20 unique runs). Augmentation ablation: standard (aug, reference, 20 seeds $S=\{2024, \dots, 2043\}$ shared with Group A) vs. no-augmentation (noaug, 20 seeds $S=\{2024, \dots, 2043\}$). The 20 noaug runs are the unique Group D contribution; the aug reference is drawn from Group A.

Group E (exploratory, 80 runs). Eight architectural ablations (E1–E8) $\times$ 10 seeds: meta-only fusion (E1), concatenation without gate (E2), additive injection without gate (E3), ReLU gate (E4), no residual injection (E5), availability signal disabled / $q=1$ always (E6), no feature dropout (E7), zero-metadata training / trainmask (E8).

Group F (multi-split absolute-performance check, 20 runs). Four alternative fold pairs (test/val: 2/3, 3/4, 7/8, 8/9) $\times$ 5 seeds, demo+anthro condition. No statistical test across the 20 runs as if independent (pre-declared); analysis unit is the fold.

The choice of 20 seeds for Group A is a principled compromise between statistical resolution and the CPU-only computational budget. The paired Wilcoxon test at $n=20$ provides a minimum attainable exact p -value of approximately 0.000002 ($= 2/2^{20} \approx 1.9 \times 10^{-6}$), enabling detection of small effects such as the demographic-only gain of $\Delta = +0.0007$ macro-AUC. Twenty seeds is above the modal practice in supervised deep learning evaluation, where three to five seeds are commonly reported [37,38], and provides adequate power for the effect sizes observed here ($d_z=0.64$ for demo–none; $d_z=1.65$ for demo+anthro–none).

Table 1 provides a complete run-accounting ledger to support reproducibility auditing. The “Unique runs” column counts runs not present in any other group; shared reference seeds are listed in the “Reference source” column.

Table 1. Run-accounting ledger for the 250-run campaign ($S_{10}=\{2024,\dots,2033\}$; $S_{20}=\{2024,\dots,2043\}$). Groups B, C, and E share 10 Group A demo+anthro seeds as their reference, yielding 20 shared descriptors in the 270-descriptor total.

Grp	Description	Seeds	Unique	n	Role
A	3 variants (none/demo/da)	S_{20} ea.	60	20	Confirmatory
B	3 meta_hid values; ref shared w/A	S_{10} ea.	30	10	Exploratory
C	4 λ_{LAUC} values; ref shared w/A	S_{10} ea.	40	10	Exploratory
D	noaug; aug ref from A	S_{20}	20	20	Exploratory
E	8 arch ablations; std ref from A	S_{10} ea.	80	10	Exploratory
F	4 alt. fold pairs, da only	5 ea.	20	5/fold	Gen. check
Total unique runs			250		
Total descriptors (incl. shared)			270		

5.6. Statistical analysis

We report macro-AUC and macro- F_1^* as mean $\pm$ standard deviation across seeds. Marginal 95% percentile bootstrap CIs are obtained by resampling the 20 seed-level values 10,000 times; paired-difference CIs are obtained by resampling the 20 seed-matched differences. Pairwise inference uses exact two-sided paired Wilcoxon signed-rank tests [39] on seed-matched variant differences, implemented with `scipy.stats.wilcoxon` (SciPy 1.16.3, `zero_method="wilcox"`, `correction=False`, exact p -values for $n \leq 25$). We report Cohen’s d_z as the main descriptive paired effect size and Hedges-corrected g_z as a small-sample companion.

BH-FDR family (Group A, confirmatory). Benjamini–Hochberg FDR control at $q=0.05$ is applied across the **pre-specified 3-test family**: the three pairwise macro-AUC contrasts in Group A (none vs. demo; none vs. demo+anthro; demo vs. demo+anthro). With $n=20$, the attainable raw two-sided p -values are discrete; the minimum attainable exact Wilcoxon p at $n=20$ is approximately 0.000002 ($= 2/2^{20} \approx 1.9 \times 10^{-6}$, achieved when all 20 signed-rank sums equal 0 or 210).

Exploratory groups (B, C, D, E). Results for Groups B, C, D, and E are exploratory. Wilcoxon p -values are reported for descriptive purposes without multiple-comparison correction; no confirmatory inference should be drawn from these values.

Secondary per-class sub-family (DA–NONE, Table 8). As a secondary, post-hoc analysis, BH-FDR correction is additionally applied within the 5-test sub-family of per-class DA–NONE Wilcoxon tests (NORM, MI, STTC, CD, HYP). This sub-family was not pre-specified and serves descriptive purposes only. Per-class tests for the demo–none contrast are exploratory and reported as raw p -values only.

Group F (multi-split absolute-performance check). No statistical test across the 20 Group F runs as if independent. Analysis is at the fold level (4 units); the primary quantity is the inter-fold macro-AUC range.

Decision thresholds are tuned per class on fold 9 and fixed for fold 10; reported p -values are therefore conditional on this validation-based model-selection protocol.

5.7. Methodological contribution and fit with Mathematics

The seed-matched paired-Wilcoxon measurement pipeline introduced here is a generalisable statistical framework for multimodal-ML ablations, and constitutes the primary methodological contribution of this work to the mathematical sciences. For each seed s and variant V_k , one trains to obtain a scalar evaluation metric $r_{k,s}$. Pairwise incremental effects are estimated as $\delta_{jk,s} = r_{j,s} - r_{k,s}$. Inference proceeds by applying the exact two-sided Wilcoxon signed-rank test (a distribution-free rank test on signed differences), BH-FDR correction over the pre-specified confirmatory test family (controlling the expected proportion of false discoveries among rejected hypotheses), and percentile bootstrap confidence intervals on the seed-matched differences. Effect sizes are reported as Cohen's d_z with Hedges' small-sample correction g_z . Exploratory groups report raw p -values descriptively without FDR correction. The framework provides a principled approach to the multiple-comparison problem in deep learning ablation studies, separating a pre-specified confirmatory family (3-test BH-FDR) from exploratory nominal p -values, and quantifying incremental contributions with calibrated uncertainty. It applies to any setting satisfying three conditions: (i) a baseline model can be paired with an augmented variant across the same random seeds; (ii) the number of seeds is small (say, $n \in [5, 20]$); and (iii) one seeks calibrated inference about the incremental contribution of an additional input modality or component.

6. Results

6.1. Aggregate performance

All reported p -values are conditional on the validation-based model-selection protocol. Table 7 reports macro-AUC and macro- F_1^* for each variant as mean $\pm$ standard deviation across 20 seeds, together with 95% bootstrap CIs and paired Wilcoxon p -values. The demo+anthro variant reaches 0.9289 ± 0.0013 macro-AUC on fold-10, numerically close to the 10-fold CV mean of the xresnet1d101 benchmark of [6] (0.928 ± 0.005 ; direct comparison is limited as our evaluation uses fold-10 alone, and Group F descriptively shows that absolute demo+anthro AUC was lower on fold 10 than on four alternative folds, consistent with fold-level difficulty variation).

A key finding is that **both demographic-only and full-metadata gains are statistically significant** across 20 paired seeds. The demo–none difference is $+0.0007 \pm 0.0002$ (SEM) macro-AUC with a 95% paired bootstrap CI of $[0.0002, 0.0011]$ ($p_{\text{raw}}=0.0094$, $p_{\text{BH}}\approx 0.009$, $d_z=0.64$, $g_z=0.61$; 16 of 20 paired seed differences positive). The demo+anthro–none contrast yields $+0.0019 \pm 0.0003$ (SEM) macro-AUC with a 95% paired bootstrap CI of $[0.0014, 0.0023]$ ($p_{\text{raw}}\approx 0.00001$, $p_{\text{BH}}<0.001$, $d_z=1.65$, $g_z=1.58$; 19 of 20 differences positive). The demographic gain is real but smaller than the anthropometric component; the incremental anthropometric step (demo+anthro–demo) is itself significant ($+0.0012 \pm 0.0003$ (SEM), $p_{\text{BH}}\approx 0.001$, $d_z=0.95$). These results should be read as protocol-specific measurements rather than as large clinical effect sizes.

6.2. Per-class decomposition

Table 8 and Figs. 3 and 5 report class-wise AUC and the corresponding Δ -AUC for the demo+anthro minus none contrast. With 20 seeds and BH-FDR correction, **three classes reach significance: NORM** ($+0.0016$, $p_{\text{BH}}<0.0001$), **MI** ($+0.0058$, $p_{\text{BH}}<0.0001$), and **STTC**

(+0.0018, $p_{\text{BH}}=0.00035$). The MI effect has the largest absolute delta (+0.0058) and a strong $d_z=2.26$. CD shows a negligible shift (-0.0003 , $p_{\text{BH}}=0.648$). **HYP shows no statistically significant gain** (+0.0004, $p_{\text{BH}}=0.648$); we return to this HYP pattern in Section 7.

For the demo—none contrast, exploratory per-class Wilcoxon tests identify significance for NORM (+0.0006, $p_{\text{raw}}=0.006$) and MI (+0.0020, $p_{\text{raw}}=0.003$), indicating that even demographic variables alone carry a per-class signal in these two morphology-rich classes. These per-class tests are not in the pre-specified BH-FDR family and are reported descriptively.

6.3. Training dynamics

Figure 2 shows the validation macro-AUC trajectory across 10 training epochs for all three variants, displayed as mean $\pm$ standard deviation across all 20 seeds. The cross-seed SD narrows from 0.008–0.010 at epoch 1 to ≈ 0.001 at epoch 10. All three variants converge to near-final performance by epoch 7–8. The final-epoch means and SDs are: none 0.9320 ± 0.0012 , demo 0.9323 ± 0.0008 , demo+anthro 0.9330 ± 0.0012 .

6.4. Missingness robustness

Figure 4 shows inference-time macro-AUC under synthetic anthropometric masking. The empirical masking curve is evaluated on the demo+anthro variant across **10 seeds** (seeds 2024–2033; the pre-declared analysis plan specified 20 seeds—this deviation was made for computational efficiency; see Limitation 8); the none and demo variant traces are included as structural reference baselines (they are immune to anthropometric masking by construction: none receives no metadata, and demo contains no anthropometric fields to mask). At each masking rate $\rho \in \{0, 0.25, 0.5, 0.75, 1.0\}$, each of the three anthropometric fields (height_z, weight_z, BMI_z) is independently zeroed with probability ρ at inference time; the three corresponding missingness indicators (m_h , m_w , m_{bmi}) and availability mask bits are updated consistently. Demographic fields (age_z, sex01) are never masked and remain available throughout. Note that BMI is derived from height and weight, so independent masking of the three fields can produce states where BMI is masked while only one of height or weight is masked; this is treated as a maximal stress test rather than a physiologically realistic dropout scenario, and the scenario was not explored under joint anthropometric-field masking. The stress-test curve remains stable: full anthropometric masking costs only ≈ 0.0014 macro-AUC relative to the unmasked model (Panel A, Δ annotation). At $\rho=1.0$ the demo+anthro model converges to the demo baseline (difference -0.00018 , less than $\frac{1}{6}$ cross-seed SD) and remains +0.00047 above the none baseline. Panel B clarifies that the none variant is immune by construction (flat), while demo shows minimal drift (< 0.0001). This demonstrates graceful degradation under deployment-time field dropout.

6.5. Seed distribution and fused-branch gain

Figure 6 shows the distribution of test macro-AUC across all 20 seeds for each variant. The demo+anthro violins are shifted upward relative to none and demo, with minimal within-variant variance ($\text{SD} \leq 0.0013$). The significance brackets confirm the two BH-FDR confirmatory contrasts (Section 5). Across seeds, the fused head consistently outperforms the ECG-only auxiliary head for demo+anthro: the $\text{AUC}_{\text{fused}} - \text{AUC}_{\text{ecg}}$ gap is positive for **all 20 of 20 seeds** (median +0.0016; Panel B of Fig. 6; note that the ECG auxiliary head is trained jointly inside the multimodal model, not as an independent ECG-only baseline—the none variant serves as the independent ECG-only reference; note also that the fused head is trained with both BCE and AUC surrogate losses, while the ECG auxiliary head is trained with BCE only, so the fused–ECG gap partly reflects this training asymmetry in addition to the metadata contribution), whereas for none and demo this gap is centred near zero (8/20 and 16/20 positive; medians -0.0001 and $+0.0003$ respectively). This is consistent with $w^*=1.0$ fixed a priori: when metadata help, the fused branch captures the gain.

6.6. Comparison with prior work

Tables 9–10 position EZNX-ATLAS-A through a comparability audit rather than within a heterogeneous leaderboard. Our direct numerical comparator is the canonical PTB-XL super-diagnostic benchmark of Strodthoff *et al.* [6]: the none variant lies approximately 0.001 macro-AUC below the 10-fold CV mean (0.9271 vs. 0.928; comparison is indicative as our evaluation uses fold-10 alone, and Group F descriptively shows that absolute macro-AUC varied across splits, with fold 10 below the inter-fold range) while using a substantially smaller trainable model. The self-supervised and multimodal papers listed in Table 10 address neighbouring questions, but their task definitions, datasets, or metadata modalities differ from our seed-matched PTB-XL demographic/anthropometric ablation.

6.7. Architectural diagnostics and sensitivity ablations

With $w^*=1.0$ fixed a priori, test predictions are those of the fused head alone for all runs.

Groups B and C: hyperparameter sensitivity (exploratory).

Table 2 reports multi-seed (10 seeds) exploratory sensitivity results for Groups B and C. Group B tests three non-reference `meta_hid` values {32, 64, 256} against the reference (128, shared with Group A); all Wilcoxon $p > 0.32$ versus the reference, spanning the macro-AUC range 0.9285–0.9296 with overlapping intervals. Note that `meta_hid` controls both the projection dimension and the gate hidden dimension simultaneously (entangled); varying it does not isolate a single architectural degree of freedom. Group C tests four non-reference λ_{LAUC} values {0.00, 0.04, 0.12, 0.16} against the reference (0.08, shared with Group A); all variants are within the noise of the reference, with p -values ranging from 0.557 to 0.695. Note that $w_{\text{fused}} = \max(0, 0.60 - \lambda_{\text{LAUC}})$, so the two loss weights change simultaneously when λ_{LAUC} is varied. These results show no detectable sensitivity under this exploratory design, rather than identifying an optimal configuration; no BH-FDR correction is applied to exploratory groups.

Table 2. Exploratory sensitivity: fusion capacity (Group B, `meta_hid` controls projection and gate hidden dim jointly) and LAUC-BCE balance (Group C; $w_{\text{fused}} = \max(0, 0.60 - \lambda_{\text{LAUC}})$, both weights change simultaneously). `demo+anthro` condition, 10 seeds. † = default value. Wilcoxon p -values ($n=10$, exact two-sided) are reported descriptively; no FDR correction is applied to exploratory groups.

Group	Parameter / Value	Macro-AUC (mean±SD)	Wilcoxon p vs. ref
B	<code>meta_hid</code> = 32	0.9285 ± 0.0013	0.322
B	<code>meta_hid</code> = 64	0.9285 ± 0.0014	0.432
B	<code>meta_hid</code> = 128† (ref)	0.9289 ± 0.0012	—
B	<code>meta_hid</code> = 256	0.9296 ± 0.0010	0.322
C	λ_{LAUC} = 0.00	0.9286 ± 0.0017	0.557
C	λ_{LAUC} = 0.04	0.9291 ± 0.0012	0.695
C	λ_{LAUC} = 0.08† (ref)	0.9289 ± 0.0012	—
C	λ_{LAUC} = 0.12	0.9288 ± 0.0008	0.625
C	λ_{LAUC} = 0.16	0.9291 ± 0.0008	0.557

Group D: augmentation ablation (exploratory).

Training without stochastic augmentation (`noaug`) yields macro-AUC 0.9275 ± 0.0012 versus 0.9289 ± 0.0012 for the augmented reference ($\Delta = -0.0014$, nominal $p = 0.002$, $r = 0.725$, $n = 20$; exploratory, no FDR correction applied). This is the only exploratory group to show a nominally significant difference from the reference configuration. The result suggests that the light augmentation block (Gaussian noise, temporal shift, amplitude jitter) contributes

a measurable positive effect and should be treated as a factor to control and report rather than an optional add-on; confirmatory inference would require a pre-declared seed budget and BH-FDR correction.

Group E: architectural ablations (exploratory).

Table 3 reports macro-AUC for eight architectural ablations against the standard configuration (10 seeds each). Two extreme variants show catastrophic degradation: **E1 meta-only** (0.6966 ± 0.0021 , $p=0.002$, exploratory) suggests that the ECG signal is indispensable under this design, and **E8 trainmask** (0.5046 ± 0.0180 , $p=0.002$)—in which the model is trained with zeroed metadata and then evaluated with real metadata—is a distribution-shift negative control: training without the metadata pathway prevents its use at inference time. This result does not imply that metadata must generally be present during training; the none variant (0.9271 macro-AUC) demonstrates that ECG-only training without any metadata pathway is fully viable. All six remaining ablations (E2–E7) show macro-AUC within ± 0.0004 of the reference with $p > 0.27$, indicating that the specific choice of fusion mechanism (concatenation, additive injection, GLU gate) and the inclusion of residual injection, availability signal, and feature dropout do not materially affect macro-AUC.

Table 3. Architecture ablations (Group E, exploratory). demo+anthro variant, 10 seeds ($\mathcal{S}_{10} = \{2024, \dots, 2033\}$). E8 (trainmask): standard architecture trained with $\mathbf{x}_{\text{meta}} = \mathbf{0}$, $\mathbf{m} = \mathbf{0}$; evaluated with real metadata. Wilcoxon p -values ($n=10$, exact two-sided) vs. Standard reference (also 10 seeds, $\mathcal{S}_{10}$; the full 20-seed $\mathcal{S}_{20}$ statistics are reported in Table 7).

Configuration	Macro-AUC (mean±SD)	Wilcoxon p
Standard (reference, Group A)	0.9289 ± 0.0012	—
E1: Meta-only fusion	0.6966 ± 0.0021	0.002
E2: Concatenation (no gate)	0.9290 ± 0.0010	0.922
E3: Additive injection (no gate)	0.9290 ± 0.0012	0.846
E4: ReLU gate (no sigmoid)	0.9288 ± 0.0014	0.770
E5: No residual injection	0.9292 ± 0.0012	0.695
E6: No availability signal ($q=1$)	0.9287 ± 0.0013	0.557
E7: No feature dropout	0.9293 ± 0.0008	0.275
E8: Trainmask (zero meta at training)	0.5046 ± 0.0180	0.002

Group F: multi-split absolute-performance check (descriptive).

Table 4 reports macro-AUC for the demo+anthro model on four alternative test folds (5 seeds each). The inter-fold range is $[0.9400, 0.9445]$, with a mean of 0.9425 across the four folds. **Fold 10 (Group A reference, 0.9289 ± 0.0013) lies below this range by 0.0111–0.0156 macro-AUC**, indicating that absolute performance varied across split definitions. Group A absolute macro-AUC for the demo+anthro condition is therefore conditioned on a more challenging split; the within-fold paired variant comparisons (none vs. demo vs. demo+anthro) in Group A are unaffected by this fold-level variation. *Important caveat:* the metadata z-scoring and imputation index was constructed from folds 1–8 of the primary split. For all four Group F alternative splits, the test folds (2, 3, 7, 8) were included in the original scaler; this normalization dependency means the absolute macro-AUC values in Table 4 are not fully independent of the primary split’s preprocessing and should be interpreted as descriptive only. Likewise, only the demo+anthro variant was evaluated on these alternative splits; the none and demo variants were not re-run, so within-fold paired gains cannot be assessed for Group F.

Table 4. Multi-split absolute-performance check (Group F, descriptive only). *demo+anthro* model only, 5 seeds per split. Analysis unit: fold (4 units); no statistical test across the 20 runs as if independent. Absolute performance varied across split definitions; fold 10 had lower absolute macro-AUC than the four alternative test folds. *Normalization caveat:* the metadata z-scoring index was built from folds 1–8 of the primary split; the Group F test folds (2, 3, 7, 8) were part of this scaler and are therefore not fully independent of the primary preprocessing—treat all values as descriptive. The *none* and *demo* variants were not re-evaluated on these splits; within-fold metadata gains cannot be assessed. Group A (fold 10) reference shown for comparison.

Test fold	Val fold	<i>n</i> seeds	Macro-AUC (mean±SD)
2	3	5	0.9422 ± 0.0007
3	4	5	0.9400 ± 0.0006
7	8	5	0.9434 ± 0.0007
8	9	5	0.9445 ± 0.0013
10 (Group A)	9	20	0.9289 ± 0.0013 (reference)
Inter-fold range (folds 2,3,7,8): [0.9400, 0.9445]; mean: 0.9425			

7. Discussion

7.1. Q1 — when do metadata help?

Our 20-seed protocol yields a clear and nuanced answer. **Both demographic and anthropometric variables show statistically significant gains.** Demographics alone (age, sex) yield $\Delta = +0.0007$ macro-AUC ($p_{\text{BH}} \approx 0.009$, $d_z = 0.64$, $n_{\text{pos}} = 16/20$ seeds). This small but reproducible effect is consistent with the partial recoverability argument of Attia *et al.* [18]: age and sex are substantially recoverable from the waveform (sex AUC 0.97, age MAE 6.9 y), yet a residual increment not fully captured by the ECG-only model remains detectable with 20 paired seeds. The incremental anthropometric gain beyond demographics is larger and more significant ($\Delta = +0.0012$, $p_{\text{BH}} \approx 0.001$, $d_z = 0.95$): body-size variables provide complementary information not easily recoverable from the 12-lead waveform alone. Both effects remain small in absolute terms ($\Delta < 0.002$ macro-AUC) and should be read as protocol-specific measurements rather than as clinically large effect sizes. No clinical decision benefit, calibration improvement, or net-reclassification gain is claimed; whether these AUC increments translate to any clinical utility requires prospective evaluation on held-out cohorts.

7.2. Q2 — which class benefits from which metadata? The MI/STTC/NORM finding

With 20 seeds and BH-FDR correction, the class-wise effect of full metadata localises to three classes. **MI** shows the strongest effect ($+0.0058$, $p_{\text{BH}} < 0.0001$, $d_z = 2.26$), followed by **STTC** ($+0.0018$, $p_{\text{BH}} = 0.00035$) and **NORM** ($+0.0016$, $p_{\text{BH}} < 0.0001$). Exploratory per-class tests for the demographic-only contrast (*demo*–*none*) also identify nominal significance for NORM ($p_{\text{raw}} = 0.006$) and MI ($p_{\text{raw}} = 0.003$); these tests are not in any pre-specified BH-FDR family and are reported descriptively. This pattern suggests that age and sex supply a useful prior for morphology-sensitive classes even when anthropometrics are absent. Plausible but non-causal hypotheses include body-size-conditioned calibration of ST-segment amplitudes, Q-wave depth, and related markers for MI, and thoracic electrical property effects for STTC; these are associations consistent with the observed patterns, not causal attributions.

7.3. The HYP paradox: no detectable gain at the superclass level

Even with 20 seeds, **HYP shows no statistically significant gain** ($+0.0004$, $p_{\text{BH}} = 0.648$). This pattern is consistent with four explanatory hypotheses: (1) low HYP prevalence limits paired test power for a genuinely small effect; (2) the TSPool’s max / min voltage statistics already partially absorb body-habitus effects; (3) superclass heterogeneity dilutes a more

specific LVH-related signal; and (4) HYP label noise in borderline cases attenuates any body-habitus signal. The non-significant HYP result with 20 seeds makes a large effect unlikely under this protocol, though smaller or subclass-specific effects (e.g., an LVH-specific signal beneath the superclass label) remain possible and were not tested here. Exploratory examination of the demo—none HYP test ($\Delta = +0.0015$, $p_{\text{raw}} = 0.048$) reveals a borderline positive demographic signal that does not pass the exploratory threshold of $p < 0.01$ and is not sustained in the full-metadata DA—NONE contrast ($p_{\text{BH}} = 0.648$), consistent with the interpretation that anthropometric variables do not amplify the marginal demographic effect for HYP.

7.4. CD: demographics and the confounding role of age

The demo variant produces a slight negative trend in CD AUC (-0.0011 , $p_{\text{raw}} = 0.123$, exploratory), but this does not survive BH-FDR correction at 20 seeds. This remains consistent with the age-as-confounder hypothesis: directly supplying age may shift the CD decision boundary, but the effect is not reproducibly negative. The demo+anthro CD delta is also non-significant (-0.0003 , $p_{\text{BH}} = 0.648$).

7.5. Q3 — graceful degradation

The availability-aware pathway—combining missingness indicators (Eq. (1)), zero-initialised residual injection (Eq. (8)), and sample meta-dropout during training—yields a model whose macro-AUC under full anthropometric suppression remains within ≈ 0.0014 of the full unmasked variant (Fig. 4), converging to the demographics-only baseline. In PTB-XL, q_{meta} saturates to 1 for nearly all records when age and sex are present (as in the demo and demo+anthro conditions); the primary graceful-degradation mechanism under anthropometric dropout is therefore **sample meta-dropout** (which trains the model to function without anthropometric inputs) and **median imputation** (which provides a neutral fallback value). The missingness indicators m_h , m_w , m_{bmi} contribute primarily when the corresponding field is present (their availability mask equals 1 when the field is observed, and 0 when the field is absent); they do not act as an explicit “field is missing” signal through the availability-gating pathway when a field is absent. q_{meta} acts mainly as a hard gate between the all-metadata and no-metadata conditions. This robustness is demonstrated under synthetic inference-time masking; generalisability to structurally missing data under MCAR, MAR, or MNAR mechanisms remains an open evaluation direction.

7.6. Group D: augmentation should be controlled and reported

The exploratory Group D result ($\Delta = -0.0014$, nominal $p = 0.002$, $n = 20$, no FDR correction applied) reveals that the light stochastic augmentation block—Gaussian noise, temporal shift, and amplitude jitter—contributes a measurable positive effect. This nominally significant finding (the only such result among all exploratory groups) suggests that augmentation should be treated as a design factor to control and report rather than an optional add-on; confirmatory inference would require a pre-declared seed budget and BH-FDR correction.

7.7. Q4 — comparability of earlier multimodal gains

Our 20-seed protocol reveals within-variant standard deviations of 0.0010–0.0013 macro-AUC, of the same order as the measured between-variant effects. The audited literature does not provide a directly comparable PTB-XL experiment separating age/sex from height/weight/BMI with repeated seeds, paired tests, and missingness stress testing. We therefore position this paper as providing a narrow but statistically honest reproducibility baseline: $+0.0007$ from demographics and $+0.0019$ from full metadata, with class-resolved and missingness-audited evidence.

7.8. Q5 — architectural ablations: ECG is essential, gating is flexible

Exploratory Group E ablations (10 seeds, uncorrected) suggest two patterns. First, the ECG signal is indispensable: the meta-only variant (E1) collapses to 0.6966 macro-AUC, far below any ECG-based variant in this study. Second, the specific fusion mechanism does not drive macro-AUC differences at this scale: all six mechanism variants (E2–E7) achieve 0.9287–0.9293, within the noise of the standard configuration ($p > 0.27$ in all cases). This is notable because E2 (simple concatenation) and E3 (additive injection) achieve performance comparable to the GLU gate, suggesting that the gate’s added complexity is not necessary to capture the metadata signal at this effect magnitude. The E8 result (0.5046 macro-AUC) is a distribution-shift negative control: training with zeroed metadata and then evaluating with real metadata induces a severe train/inference distribution mismatch. This does not imply that metadata must generally be present during training; the none variant (0.9271 macro-AUC) demonstrates that ECG-only training without any metadata pathway remains fully competitive.

7.9. Q6 — fold-level variation: absolute performance varied across split definitions

Group F shows that the four alternative fold pairs (folds 2, 3, 7, 8 as test folds) all achieve macro-AUC in the range [0.9400, 0.9445], substantially above fold 10 (0.9289). This 0.011–0.016 gap is consistent with fold-level difficulty variation in stratified k-fold settings and indicates that absolute demo+anthro macro-AUC varied across the tested splits. The within-fold paired comparisons of Group A are unaffected by this fold-level variation; the absolute difference should be noted when positioning fold-10 results relative to benchmarks evaluated on averaged or best-of-fold metrics. Two important limitations apply to Group F: first, the metadata normalization index was built from folds 1–8 of the primary split, meaning the alternative test folds (2, 3, 7, 8) were included in the original scaler—these values should therefore be treated as descriptive rather than confirming a pattern independent of the primary preprocessing; second, because none and demo variants were not re-evaluated on the alternative splits, within-fold metadata gains cannot be assessed.

7.10. Sensitivity ablations: no detectable sensitivity to hyperparameters

The exploratory multi-seed sensitivity analyses (Groups B and C, Section 6.7) are consistent with the reported results not being sensitive to a specific metadata hidden size or AUC-surrogate weight. All seven non-reference variants (three metaH, four lauc) show Wilcoxon $p > 0.32$ against the standard configuration, with macro-AUC ranges spanning less than 0.0012; no confirmatory inference is drawn from these exploratory results.

7.11. Why small measured effects remain informative

Reporting +0.0007 and +0.0019 macro-AUC gains is a refinement of the multimodal claim rather than a null result. The paired seed-level 95% CIs are strictly positive for the full-metadata contrast ([0.0014, 0.0023]), supporting a reproducible directional effect without implying a large clinical effect size. The demographic-only CI ([0.0002, 0.0011]) is positive but narrow, consistent with a small real effect that required 20 seeds to detect. These results indicate that: (i) a strong ECG-only model is already close to the incremental information supplied by the tested metadata; (ii) the residual signal localises to morphology-rich classes (MI, STTC, NORM); (iii) demographic fusion alone can create class-specific shifts audible only through paired multi-seed statistics; and (iv) anthropometrics contribute a second, statistically significant increment beyond demographics.

7.12. Limitations

1. **Evaluation scope and split-level variation.** The confirmatory Group A results are conditioned on fold 10; Group F descriptively shows that absolute demo+anthro macro-AUC varied across the tested splits, with fold 10 lying below the inter-fold range [0.9400, 0.9445] by approximately 0.011–0.016. The within-fold paired variant comparisons in Group A remain valid and are unaffected by this absolute-value difference. Note that Group F has a normalization dependency (see Limitation 7), so this variation should be interpreted cautiously. No external cohort evaluation has been performed, limiting generalisation across institutions, acquisition devices, demographic distributions, and temporal shifts.
2. **Task and metadata scope.** The analysis is restricted to the 5-superclass task and to age, sex, height, weight, and BMI. Variables such as blood pressure, electrolytes, and echocardiographic measurements are absent. Age, sex, and body size are not purely biological descriptors—they also carry population-structure signal—so measured gains cannot be unambiguously attributed to physiological mechanisms alone.
3. **Model selection and downstream metrics.** Class-wise F_1 thresholds and checkpoint selection are all determined on fold 9 and fixed for fold 10. The reported p -values are therefore conditional on this validation-based selection protocol, and F_1 should be read more cautiously than AUC.
4. **Architectural attribution.** Although Group E ablations (E2–E7) show that individual components do not materially affect macro-AUC, the entangled nature of certain hyperparameters (e.g., `meta_hid` in Group B controls both projection and gate dimensions) prevents clean attribution. The metadata MLPs retain BatchNorm, which may be less stable than layer-normalised alternatives when metadata-rich samples are rare within a mini-batch.
5. **Statistical granularity.** The reported bootstrap intervals quantify variability across seed-level runs rather than patient-level resampling, and the seed-matched comparisons share the same dataset, preprocessing pipeline, and split definition. The paired contrasts are therefore correlated repeated runs on one fixed cohort split rather than independent cohort replications. Exploratory group results (B, C, D, E) report raw p -values without FDR correction and should not support confirmatory conclusions.
6. **Reproducibility provenance.** All 250 runs were executed on a CPU-only machine (Intel Core i5, 8 GB RAM) using PyTorch 2.3.1+cpu in a local Jupyter Notebook environment under Windows. CPU-only execution eliminates one source of non-determinism (CUDA parallel atomic operations); results may nonetheless differ across different CPU architectures, BLAS library versions, or threading configurations, so identical hardware and software environment is required for bitwise reproduction. The ECG signal division by 5.0 is applied in the training collate functions (`collate_fn_augmented` and `collate_fn_val`) rather than in the dataset loader; the source snapshot loader applies this division standalone for self-consistency.
7. **Group F normalization dependency and protocol deviation.** The metadata z-scoring and imputation index was constructed once from folds 1–8 of the primary split. For all four Group F alternative splits, the respective test folds (2, 3, 7, 8) were part of the original scaler, introducing a normalization dependency. Group F absolute macro-AUC values are therefore descriptive only and should not be interpreted as fully independent fold-robustness evidence. Recalculating the scaler per alternative split would require re-running the full index construction pipeline. *Note:* Group F was pre-declared as “confirmatory (generalisation check)” in the analysis plan; given the normalization dependency described above, we downgrade its interpretation to descriptive. This constitutes a protocol deviation from the pre-declared plan.

8. **Pre-specified analyses not included in the main text.** The pre-declared analysis plan (SHA-256: 4ad22e90 . . . aea5c3dc) specifies several analyses not reported in the main manuscript: (a) classical baselines (logistic regression and XGBoost on metadata features); (b) post-hoc metadata decomposition controls (shuffle_val, mask_only, and none inference conditions applied to the 60 Group A checkpoints without retraining); (c) per-class AUPRC; (d) calibration metrics (Brier score, ECE); and (e) subgroup AUC by age tertile, metadata completeness, and sex. Additionally, the missingness robustness curve (Section 6) was specified over 20 seeds but evaluated on 10 seeds for computational efficiency—this constitutes a minor protocol deviation. These analyses were defined in the plan prior to any training run. Their omission from the main manuscript reflects scope decisions made before examining results; results and/or justifications are available in the public repository (https://github.com/ezyneane/ezyx-atlas-a_github_code).
9. **Calibration and clinical utility.** No calibration metrics (ECE, Brier score) or decision-curve analysis are reported. The findings are restricted to discrimination (macro-AUC) and should not be interpreted as evidence of deployable clinical utility without calibration evaluation, net-reclassification analysis, and prospective or external cohort validation.

8. Conclusions

EZNX-ATLAS-A reaches macro-AUC comparable to the canonical PTB-XL superclass benchmark while serving as an instrument for measuring the incremental contribution of clinical metadata across a **250-run campaign organised in six experimental groups (A–F)**. The main findings are:

- (a) **Demographic variables** (age, sex) **add a small but statistically significant gain** ($\Delta = +0.0007$ macro-AUC, 95% paired seed-level CI [0.0002, 0.0011], $p_{BH} \approx 0.009$); the effect localises to NORM and MI and is consistent with a residual demographic signal not fully recovered from the waveform alone.
- (b) **Anthropometric variables** (height, weight, BMI) **provide an additional highly significant gain** of +0.0012 over demographics ($p_{BH} \approx 0.001$), for a total full-metadata gain of +0.0019 over ECG alone ($p_{BH} < 0.001$, 95% CI [0.0014, 0.0023]); the class-wise signal concentrates in MI, STTC, and NORM.
- (c) **HYP shows no significant effect** under this superclass protocol ($p_{BH} = 0.648$), consistent with voltage-based LVH criteria being diluted by superclass heterogeneity, and **CD shows no significant effect** ($p_{BH} = 0.648$), despite the demographic CD hypothesis.
- (d) The **availability-aware pathway**—combining missingness indicators, zero-initialised residual injection, and sample meta-dropout—remains stable under simulated anthropometric missingness, losing only ≈ 0.0014 macro-AUC under full anthropometric masking and converging to the demographics-only baseline. Note that q_{meta} acts primarily as an all-or-no-metadata gate in PTB-XL; the observed robustness is mainly attributable to sample meta-dropout and median imputation (missingness indicator availability masks are 0 when fields are absent, so their primary contribution is when fields are present; see Section 7).
- (e) **Exploratory Groups B and C** (hyperparameter sensitivity, 70 unique runs) show no detectable sensitivity of macro-AUC to metadata hidden size or AUC-surrogate weight under this exploratory design (all Wilcoxon $p > 0.32$, 10 seeds each, no FDR correction applied).
- (f) **Group D** (augmentation ablation, 20 unique runs, $n=20$ seed pairs, exploratory) finds a nominally significant macro-AUC drop when augmentation is removed ($\Delta = -0.0014$, nominal $p = 0.002$, no FDR correction applied), motivating explicit con-

- trol and reporting of augmentation choices; confirmatory inference would require a pre-declared seed budget and BH-FDR correction.
- (g) **Group E** (architectural ablations, 80 runs, exploratory) suggests that the ECG signal is indispensable (meta-only AUC 0.6966) and that training with zeroed metadata and evaluating with real metadata induces severe distribution mismatch (E8 negative control, AUC 0.5046); training without any metadata pathway remains fully viable (none, AUC 0.9271). The specific fusion mechanism does not materially affect macro-AUC.
- (h) **Group F** (multi-split check, 20 runs, descriptive only) shows that absolute demo+anthro macro-AUC varied across four alternative splits (range [0.9400, 0.9445]), with fold 10 below this range; within-fold paired comparisons of Group A are unaffected. Note that the metadata normalization index was built from folds 1–8 and the alternative test folds overlap with this set, so Group F absolute values should be treated as descriptive only.

More broadly, the seed-matched paired-Wilcoxon + BH-FDR (over the pre-declared 3-test confirmatory family) + paired-bootstrap pipeline described in §5 constitutes a reusable methodological baseline for any multimodal-ML ablation in which an additional input modality is evaluated across repeated runs. The 20-seed design for Group A provides enough paired runs to detect small effects such as the demographic gain ($d_z=0.64$), highlighting the importance of adequate seed budgets in multimodal deep learning evaluation. Beyond ECG diagnostics, the seed-matched measurement framework presented here constitutes a transferable and computationally accessible statistical toolbox for quantifying incremental modality contributions with calibrated uncertainty in any multimodal-ML ablation study.

Supplementary Materials: Supplementary materials available in the public GitHub repository (https://github.com/ezynsegnane/ezyx-atlas-a_github_code) include: **Table S1** — the 3-test BH-FDR pre-specified family (Group A pairwise macro-AUC contrasts) with raw and adjusted p -values; seed-level JSON outputs for all 250 runs across six groups; seed-level CSV/Markdown statistical tables; dataset-integrity and statistical-protocol notes; environment files; and source/figure-generation scripts.

Author Contributions: Conceptualization, E.S., M.M., and B.M.A.; methodology, E.S., Y.O., K.E.W., M.C.T., and D.H.P.-O.; software, E.S.; validation, E.S., Y.O. and M.C.T.; formal analysis, E.S., K.E.W. and M.C.T.; investigation, E.S.; resources, E.S.; data curation, E.S.; writing—original draft preparation, E.S.; writing—review and editing, E.S., Y.O., K.E.W., M.C.T., M.M., D.H.P.-O., and B.M.A.; visualization, E.S.; supervision, Y.O., K.E.W., M.C.T., and D.H.P.-O.; project administration, E.S. All authors have read and approved the final version of the manuscript as submitted.

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Institutional Review Board Statement: Ethical review and institutional approval were waived for this study because it uses only the public, fully de-identified PTB-XL dataset [5], whose original collection was approved by the institutional ethics committee of PTB Berlin.

Informed Consent Statement: Patient consent is not required for this secondary analysis of a public, fully de-identified dataset. Informed consent for the original PTB-XL data collection was obtained according to the PTB Berlin ethics protocol [5].

Data Availability Statement: The PTB-XL v1.0.3 dataset is freely available on PhysioNet (<https://physionet.org/content/ptb-xl/1.0.3/>, doi:10.13026/kfzx-aw45; accessed 19 May 2026) [5,36,40]. The complete training code, all 250 seed-level result JSON files across six experimental groups, derived Group A statistical artifacts regenerated from all 60 seed JSONs (20 seeds $\times$ 3 variants), and the six manuscript figures are publicly available at https://github.com/ezynsegnane/ezyx-atlas-a_github_code (MIT licence; commit ebc1e5e anchors the pre-declared analysis plan; a versioned tag or Zenodo

DOI will be created upon acceptance to provide a fixed reference). The pre-declared analysis plan (analysis_plan.md) is included in the public repository; its SHA-256 is 4ad22e90...aea5c3dc (full hash in each run JSON under key analysis_plan_sha256), matching the plan committed before any training run (commit ebc1e5e). The public missingness report covers the demo+anthro variant only (10 seeds, 2024–2033); none and demo masking curves are structural reference baselines immune to anthropometric masking by construction and are shown in Fig. 4 at their 20-seed means from Group A (the 10 vs. 20-seed distinction does not affect their masking-rate traces). Patient-level probability arrays (per-patient class probabilities, labels, and split identifiers) were not individually archived due to storage constraints; the seed-level JSON files provide aggregate AUC, F1, and per-class results sufficient to reproduce the reported statistics.

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Use of Generative AI: During the preparation of this submission, the authors used Claude (Anthropic) to audit the reference list for dataset and terminology consistency, to verify manuscript compliance with journal formatting requirements, to assist with language revision and self-review of manuscript text, and to correct grammar, spelling, and punctuation errors. The authors reviewed all AI-generated suggestions and take full responsibility for the content of this publication.

Conflicts of Interest: No conflict of interest is declared by any of the authors.

Abbreviations used in this manuscript:

AUC	Area under the ROC curve
BCE	Binary cross-entropy
BH	Benjamini–Hochberg
BMI	Body mass index
CD	Conduction disturbance (PTB-XL superclass)
ECG	Electrocardiogram
EHR	Electronic health record
FDR	False discovery rate
GLU	Gated Linear Unit
HYP	Hypertrophy (PTB-XL superclass)
IQR	Interquartile range
LVH	Left-ventricular hypertrophy
MAR	Missing at random
MCAR	Missing completely at random
MI	Myocardial infarction (PTB-XL superclass)
MLP	Multilayer perceptron
MNAR	Missing not at random
NORM	Normal ECG (PTB-XL superclass)
PTB-XL	Physikalisch-Technische Bundesanstalt eXtra Large ECG corpus
SCP	Standard Communications Protocol (for ECG coding, ISO 11073-91064)
STTC	ST/T change (PTB-XL superclass)

Table 5. PTB-XL 1.0.3 dataset statistics used in this study. Superclass prevalence is the fraction of records carrying that superclass (multi-label; rows do not sum to 100%). Positive weights are derived via Eq. (2) from training-fold prevalences.

	NORM	MI	STTC	CD	HYP
Training prevalence (f_c , %)	43.6	25.1	24.0	22.4	12.2
Positive weight w_c^+	1.29	2.98	3.16	3.46	7.22
Validation prevalence (%)	43.7	24.7	24.2	22.7	12.3
Test prevalence (%)	43.8	25.0	23.7	22.6	11.9

Training / validation / test records (folds 1–8 / 9 / 10)	17,418 / 2,183 / 2,198
Metadata completeness (age / sex / height / weight / BMI)	98.7 / 100.0 / 31.9 / 43.2 / 30.9 %

Table 6. Hyperparameters and compute cost. All values are held constant across the 60 Group A runs (20 seeds $\times$ 3 variants) unless explicitly varied in an exploratory group (see Table 1). All runs were executed on a CPU-only machine (Intel Core i5, 8 GB RAM, PyTorch 2.3.1+cpu) with no GPU or CUDA device.

Hyperparameter	Value
Optimizer	AdamW
Learning rate	10^{-3}
Weight decay	5×10^{-4}
LR schedule	CosineAnnealingWarmRestarts, $T_0=10$, $T_{\text{mult}}=2$
Batch size (effective)	$32 \times 2 = 64$
Gradient clipping	maximum norm 1.0
Epochs	10
Loss weights (fused / ECG / meta / AUC surrogate)	0.52 / 0.30 / 0.10 / 0.08 (reference; Group C varies λ_{LAUC} ; effective total <1.00 when $q_{\text{meta}} < 1$)
Sample meta-dropout p	0.10 (training only)
Inference blend w^*	Fixed at 1.00 (a priori, all runs)
Parameters (trainable)	3.95 M
Training-device provenance	Intel Core i5 CPU, 8 GB RAM; PyTorch 2.3.1+cpu; CPU-only, no CUDA/GPU
Determinism flags	Code includes CuDNN deterministic/benchmark flags (inert on CPU-only); CPU execution eliminates CUDA non-determinism.
Wall-clock time per run (Group A)	43.8 min median (48.2 min mean; 43.1 to 303.5 min range)
Total compute (Group A, 60 runs)	48.2 h
Total compute (all 250 runs)	≈ 185 h

Table 7. Primary confirmatory results (Group A) on the PTB-XL fold-10 test set. Each variant was trained from **20 different random seeds**; macro-AUC and macro- F_1^* are reported as mean $\pm$ SD with 95% percentile bootstrap CIs obtained by resampling the 20 seed indices 10,000 times. For each pairwise contrast: average within-seed difference $\pm$ SEM (standard error of the mean over 20 paired differences), 95% bootstrap CI, two-sided exact Wilcoxon signed-rank p , BH-FDR-adjusted p over the **pre-specified 3-test confirmatory family**, Cohen's d_z , and Hedges g_z . Note: for the least-significant test (demo–none), $p_{\text{BH}} = p_{\text{raw}}$ because it occupies rank 3/3 in the sorted BH family.

Variant	Macro-AUC	Macro- F_1^*	95% CI / diff. CI
none	0.9271 $\pm$ 0.0011	0.7441 $\pm$ 0.0030	[0.9266, 0.9276]
demo	0.9277 $\pm$ 0.0010	0.7458 $\pm$ 0.0028	[0.9272, 0.9282]
demo+anthro	0.9289 $\pm$ 0.0013	0.7462 $\pm$ 0.0034	[0.9283, 0.9295]
Paired comparisons (macro-AUC)			
demo – none	+0.0007 $\pm$ 0.0002	$p_{\text{raw}}=0.0094$, $p_{\text{BH}}\approx 0.009$, $d_z=0.64$, $g_z=0.61$	diff. CI [0.0002, 0.0011]
demo+anthro – demo	+0.0012 $\pm$ 0.0003	$p_{\text{raw}}=0.0006$, $p_{\text{BH}}\approx 0.001$, $d_z=0.95$, $g_z=0.91$	diff. CI [0.0007, 0.0018]
demo+anthro – none	+0.0019 $\pm$ 0.0003	$p_{\text{raw}}\approx 0.00001$, $p_{\text{BH}} < 0.001$, $d_z=1.65$, $g_z=1.58$	diff. CI [0.0014, 0.0023]

Table 8. Per-class AUC on PTB-XL fold-10 (Group A), mean $\pm$ standard deviation across **20 seeds**, and paired Wilcoxon p -values for $\Delta_{\text{AUC}} = \text{demo+anthro} - \text{none}$. BH-FDR correction is applied within the 5-test per-class DA–NONE sub-family (secondary analysis, not pre-specified); bold entries reach significance. NORM, MI, and STTC reach significance (all $p_{\text{BH}} < 0.001$). CD and HYP share the same BH-adjusted p (0.648) because the BH step-down monotonicity rule reduces HYP’s initial adjustment (0.745) to match CD’s.

Variant	NORM	MI	STTC	CD	HYP
none	0.9481 $\pm$ 0.0013	0.9245 $\pm$ 0.0021	0.9355 $\pm$ 0.0014	0.9179 $\pm$ 0.0021	0.9093 $\pm$ 0.0023
demo	0.9486 $\pm$ 0.0011	0.9265 $\pm$ 0.0025	0.9357 $\pm$ 0.0015	0.9169 $\pm$ 0.0022	0.9108 $\pm$ 0.0024
demo+anthro	0.9497 $\pm$ 0.0013	0.9303 $\pm$ 0.0028	0.9373 $\pm$ 0.0014	0.9177 $\pm$ 0.0019	0.9097 $\pm$ 0.0036
$\Delta_{\text{AUC}} (\text{demo+anthro} - \text{none})$	+0.0016	+0.0058	+0.0018	−0.0003	+0.0004
Wilcoxon p -value (raw)	<0.001	<0.001	0.0002	0.648	0.596
BH-FDR p -value	<0.0001	<0.0001	0.00035	0.648	0.648
Significant after BH-FDR	yes	yes	yes	no	no

Table 9. PTB-XL-related comparators and adjacent context. Strodthoff *et al.* [6] is the direct ECG-only benchmark reference; Atwa *et al.* [13] and Holloway *et al.* [14] address related but non-identical task structures (different class sets, auxiliary inputs, or fold protocols) and serve as adjacent context rather than direct numerical comparators.

Ref.	Audited setting	Metadata / fusion	Use in this manuscript
[6]	PTB-XL super-diagnostic benchmark, official folds	Waveform-only models	ECG-only reference: xresnet1d101 achieves 10-fold CV macro-AUC 0.928 $\pm$ 0.005; our none variant scores 0.9271 on fold-10 alone. Comparison is indicative: Group F shows fold-10 had lower absolute demo+anthro AUC than the four tested alternative folds.
[13]	PTB-X/PTB-XL arrhythmia classification with 2, 5, 10, and 15 classes	Raw ECG with age/gender branches in CNN/VGG variants	The closest published PTB-XL paper to include demographic branches; however, it reports accuracy-based multi-granularity metrics on a different task structure and provides no anthropometric ablation or missingness test.
[14]	Raw-signal, feature-only, and combined models for PTB-XL multi-label diagnosis	PTB-XL+ tables fused with raw signal	Useful PTB-XL multi-label context; the auxiliary inputs are ECG-derived signal features, not structured clinical anthropometrics.
This work	PTB-XL super-diagnostic task, official folds, 20 seeds per variant	none vs. demo vs. full metadata, with availability masks	Demographics yield +0.0007 macro-AUC (localised in NORM and MI) and full metadata yield +0.0019 macro-AUC (concentrated in NORM, MI, and STTC) over ECG alone; both BH-significant. The availability-aware pathway remains robust under inference-time anthropometric dropout.

Table 10. Adjacent works whose task, dataset, or metadata definition differs from the present PTB-XL demographic/anthropometric ablation. Listed for context; direct numerical comparison is not appropriate.

Ref.	Audited setting	Metadata / fusion	Use in this manuscript
[21]	PTB-XL+ resource and associated benchmark	Uni-G, 12SL, and ECGDeli tables	Relevant same-corpus context, but the added inputs are derived from the ECG rather than from structured clinical variables.
[7–10,20]	Self-supervised or foundation ECG pretraining with PTB-XL evaluations	Waveform-only pretraining / fine-tuning	Self-supervised objectives on raw ECG waveforms; no clinical metadata involved.
[11]	AF-recurrence prediction after ablation on a non-PTB-XL cohort	Surface ECG + clinical variables	Useful multimodal example, but outside PTB-XL super-diagnostic classification.
[12]	rECGnition_v2.0 arrhythmia classification on MIT-BIH, INCARTDB, and the European ST-T Database	Patient fusion via self-attentive canonical correlation	Methodological fusion context, without a PTB-XL metadata comparator.

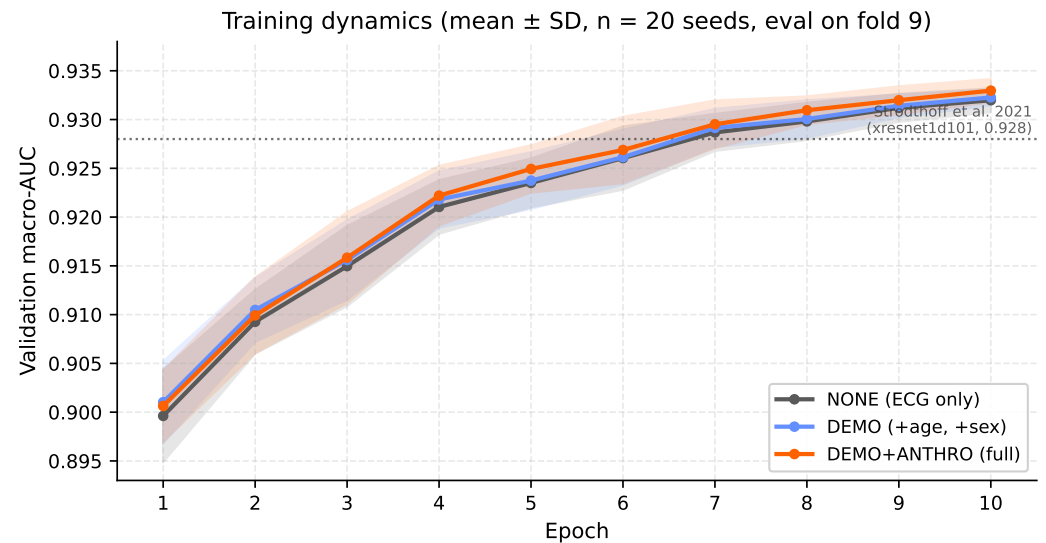


Figure 2. Validation macro-AUC trajectory across 10 training epochs, displayed as mean $\pm$ standard deviation across all **20 seeds** per variant (shaded band = ± 1 SD; evaluated on fold 9). Cross-seed SD narrows from 0.008–0.010 at epoch 1 to ≈ 0.001 at epoch 10. Final-epoch means and SDs: none 0.9320 ± 0.0012 , demo 0.9323 ± 0.0008 , demo+anthro 0.9330 ± 0.0012 . The inter-variant gap remains within one cross-seed SD throughout, consistent with the finding that the $+0.0019$ macro-AUC effect is only detectable through seed-level inference.

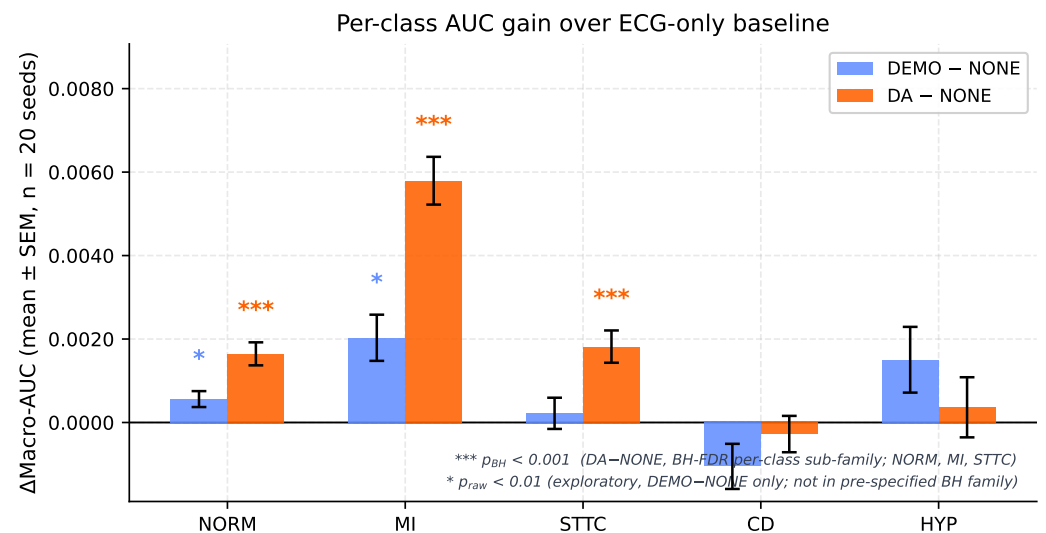


Figure 3. Per-class Δ -AUC of demo and demo+anthro relative to none, mean $\pm$ paired SEM across **20 seeds**. **Orange** markers (DA–none): *** = $p_{\text{BH}} < 0.001$ (BH-FDR corrected within the per-class DA–NONE sub-family; all three significant classes NORM, MI, STTC reach $p_{\text{BH}} < 0.001$). **Blue** markers (DEMO–none): * = $p_{\text{raw}} < 0.01$ (exploratory, not in the pre-specified BH family; NORM $p = 0.006$, MI $p = 0.003$). CD and HYP show no significant gain in either contrast.

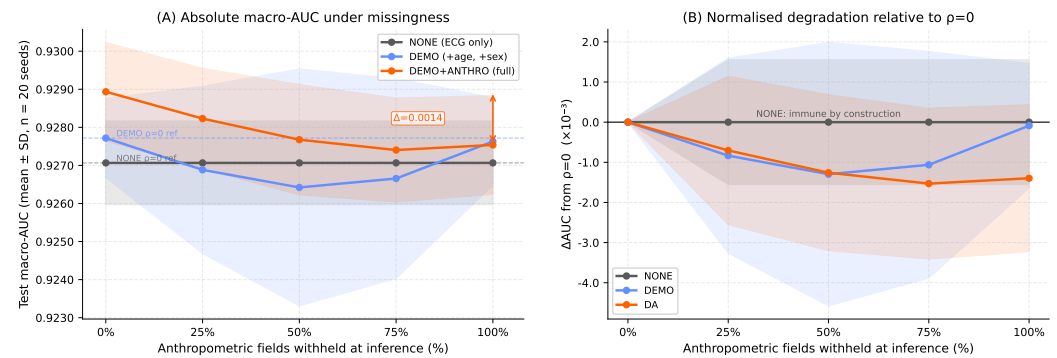


Figure 4. Missingness robustness of the availability-aware fusion pathway. The demo+anthro empirical masking curve is evaluated across **10 seeds** (seeds 2024–2033); none and demo traces are structural reference baselines (immune to anthropometric masking by construction) shown at their 20-seed mean. **(A)** Test macro-AUC as a function of the fraction ρ of anthropometric fields withheld at inference (MCAR, independently per field); dashed horizontal lines show each variant's $\rho=0$ baseline; the double-headed arrow quantifies the degradation cost. **(B)** Normalised degradation Δ AUC = $AUC(\rho) - AUC(0)$ in units of 10^{-3} for direct cross-variant comparison. Full masking ($\rho=1.0$) costs ≈ 0.0014 macro-AUC for demo+anthro, converging to the demo baseline (difference -0.00018 , within $\frac{1}{6}$ SD); the none variant is immune by construction. Availability-mask updating is performed consistently at each rate.

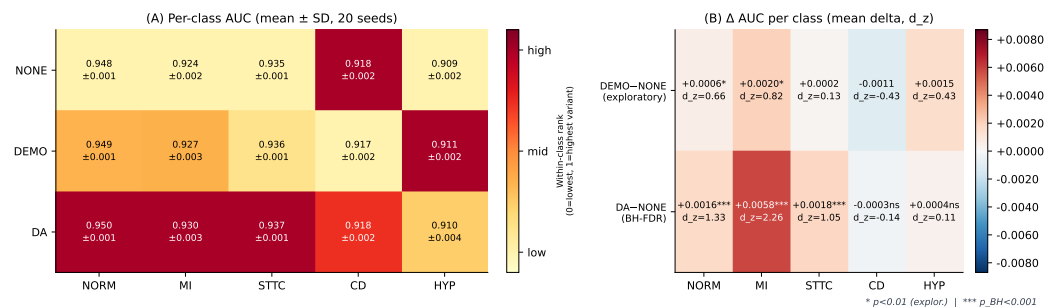


Figure 5. Per-class performance decomposition across **20 seeds**. **(A)** Per-class AUC for the three ablation variants (mean $\pm$ SD); colormap is normalised within each class column so that within-class variant differences are perceptually visible. **(B)** Δ AUC delta matrix: top row = demo–none (exploratory, threshold $p < 0.01$); bottom row = demo+anthro–none (BH-FDR corrected within per-class sub-family). Each cell shows Δ mean and Cohen's d_z . MI shows the largest absolute shift ($+0.0058$, $d_z=2.26$, $p_{BH} < 0.0001$). NORM and MI already reach significance under demographics alone (top row, $p < 0.01$); STTC additionally reaches significance only with full anthropometrics (bottom row, $p_{BH}=0.00035$). CD and HYP show no significant gain in either contrast.

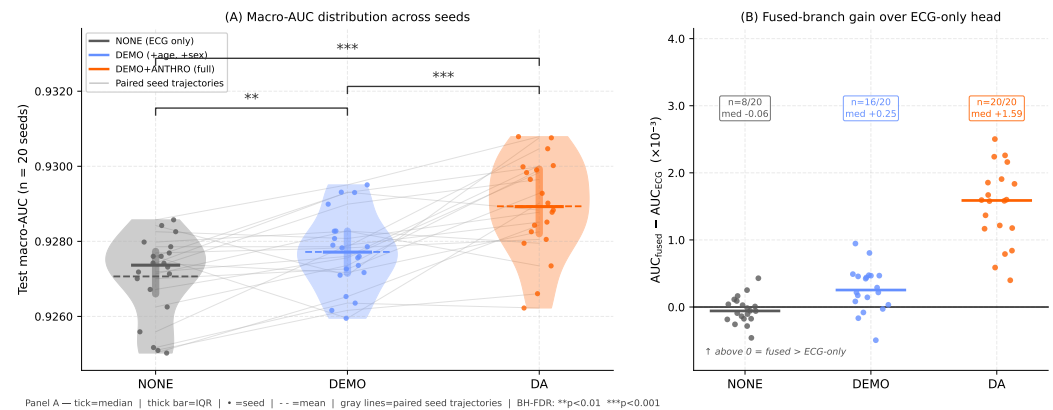


Figure 6. Multi-seed reproducibility and architectural validation across **20 seeds**. **(A)** Macro-AUC distribution per variant (violin = kernel density; thick bar = IQR; horizontal tick = median; dashed line = mean; dots = individual seeds; gray lines = paired seed trajectories). All three BH-FDR confirmatory brackets are shown: demo–none ($p_{BH} \approx 0.009$, **), demo+anthro–demo ($p_{BH} \approx 0.001$, ***), demo+anthro–none ($p_{BH} < 0.001$, ***); significance key: ** $p_{BH} < 0.01$; *** $p_{BH} < 0.001$. Gray paired trajectories show that demo+anthro exceeds none in 19 of 20 seeds and demo in 16 of 20. **(B)** $AUC_{fused} - AUC_{ECG}$ per seed: the fused branch exceeds the ECG-only auxiliary head for all 20/20 seeds under demo+anthro (median +0.0016), confirming that the fusion mechanism captures the metadata signal at inference time. The gap is near-zero for none (8/20, median −0.0001) and intermediate for demo (16/20, median +0.0003). Note: the fused head is trained with both BCE and AUC surrogate losses while the ECG auxiliary head is trained with BCE only; this training asymmetry may also contribute to the fused–ECG gap, in addition to the metadata contribution.

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